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Potential mechanisms underlying bleeding during infection with hemorrhagic fever viruses

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Abstract

Viral hemorrhagic fever (VHF) describes different diseases caused by several viruses from 6 virus families: *Filoviridae, Nairoviridae, Phenuiviridae, Hanaviridae, Arenaviridae*, and *Flaviviridae*. VHF was once considered a geographically localized problem, but due to expanding vector ranges and increased human contact with animal reservoirs and hosts the number of VHF cases is increasing. As the name indicates, VHF is associated with bleeding. Both direct effects from viral infection of host cells and indirect effects caused by the host response to the virus contribute to the dysregulation of the hemostatic system. Many studies have measured different parameters and various biomarkers in samples from infected humans and non-human primate models. For example, Ebola virus infection in a non-human primate model leads to increased tissue factor expression in peripheral blood mononuclear cells and extracellular vesicles. In dengue virus infection, thrombocytopenia and platelet dysfunction occur. There are likely both common and distinct mechanisms underlying bleeding in different VHFs, as sites of bleeding differ between the viruses. Herein, we discuss the potential mechanisms leading to bleeding during VHF, which include a consumptive coagulopathy, decreased coagulation factor production, thrombocytopenia and platelet dysfunction, and endothelial cell activation and damage resulting in increased vascular permeability. While a significant body of work exists examining different aspects of the various viral infections that may lead to bleeding, there are still many open questions and areas for investigation. Therefore, more studies are needed to better understand the mechanisms underlying bleeding in VHF caused by different viruses.

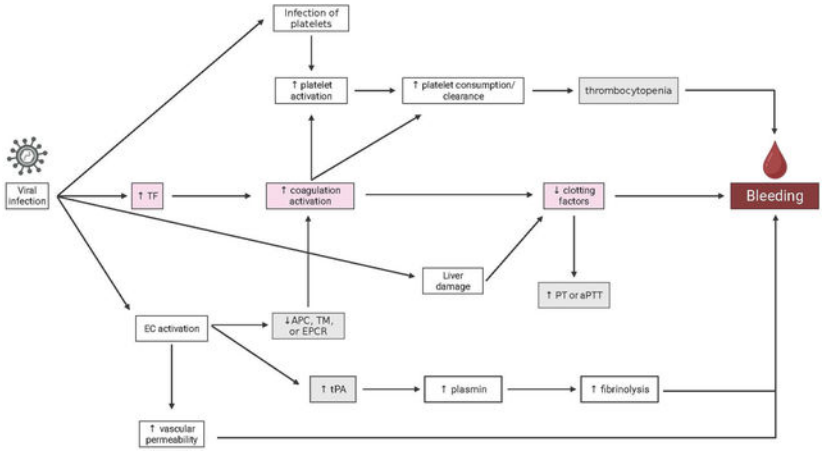
Graphical Abstract

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Disclosures

None.

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Section 1: Hemorrhagic fever viruses

Viral hemorrhagic fever (VHF) is a catch-all term for the disease induced by viruses designated as hemorrhagic fever viruses (HFVs). VHF generally begins with non-specific, febrile symptoms that may progress to more severe disease, including hemorrhagic symptoms, with a variable clinical course. While all HFVs can induce hemorrhagic symptoms, including coagulopathy, thrombocytopenia, and bleeding, they do so to differing frequencies, severity, and possibly by different mechanisms. HFVs are from 6 virus families: *Filoviridae, Nairoviridae, Phenuiviridae, Hanaviridae, Arenaviridae*, and *Flaviviridae* (Figure 1). This review will discuss Ebola virus (EBOV), Marburg virus (MARV), Lassa fever virus (LASV), Rift Valley fever virus (RVFV), Crimean Congo hemorrhagic fever virus (CCHFV), Hantaan virus (HTNV), sin nombre virus (SNV), yellow fever virus (YFV), and dengue virus (DENV). All of these viruses have small (<20 kilobases) RNA genomes. A brief overview of the virological information for each virus is presented in Table 1. The case presentation information is presented in Table 2. Importantly, the case presentation information for many of the viruses (e.g., number of infections, number of cases presenting with severe and/or hemorrhagic manifestations, and case fatality rate) is derived primarily from data on individuals presenting to a clinical setting or from epidemiological estimates. Notably, many individuals infected with some of these viruses experience sub-clinical disease or do not present to a clinic. Thus, these numbers should be considered as estimates based off of available data. The geographic distribution of the different HFVs is shown in Figure 2.

Filoviridae: Ebola virus and Marburg virus

EBOV belongs to the genus *Ebolavirus* that has 5 other species. Not all members of the *Ebolavirus* genus cause disease in humans. For those that do, the disease is collectively called Ebola virus disease (EVD) and has a pooled case fatality rate (CFR) ~60 percent (Table 2). (24) Most individuals with EVD are symptomatic and experience severe disease (Table 2). (32) EVD can be separated into 3 phases: early, peak, and resolution. The early phase begins with febrile symptoms (e.g., arthralgia, headache, fatigue, rash) that progress to severe gastrointestinal symptoms (e.g., nausea, vomiting, diarrhea). During the early phase,

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hemorrhagic symptoms (e.g., ecchymoses, bloody vomit or stool, bloody gums, conjunctival injection, nosebleeds, and bruising) may begin (Table 2). Hemorrhagic symptoms may occur in up to 50 percent of cases, but the incidence is highly variable (Table 2). (37) During the peak phase, about 4–12 days after disease onset, viral load begins to peak. With an increase in viral load, symptoms, such as gastrointestinal and hemorrhagic symptoms, continue and may worsen. Additionally, multi-organ dysfunction may be observed, including renal failure, respiratory failure, neurologic symptoms (e.g., meningoencephalitis and stroke), and cardiac problems (e.g., myocarditis and pericarditis). (37) Individuals surviving EVD enter the resolution or convalescent phase and are more susceptible to secondary infections leading to sepsis. A significant proportion of surviving individuals also report long-term sequalae, including arthralgias, myalgias, visual and auditory changes, extreme fatigue, rashes, and gastrointestinal issues. The cause of such symptoms is unknown, but these secondary complications are debilitating for affected individuals. (44)

MARV is 1 of 2 species of the genus *Marburgvirus*. MARV causes Marburg virus disease (MVD), which is separated into early, peak, and resolution phases and has a similar clinical course to EVD in terms of symptoms and overall timing. (3) There is no evidence of asymptomatic MARV disease currently. Similar to EBOV, MVD has a high proportion of cases with hemorrhagic manifestations (34–83 percent) and an average CFR of 50 percent (Table 2). (25, 28, 38)

Arenaviridae: Lassa fever virus

LASV belongs to the genus *Mammarenavirus*. There are 41 members of this genus, and only 8 of the members cause disease in humans, including LASV. (4) LASV is transmitted to humans primarily through contact with secretions from infected rodents (Table 1)(Figure 2). (17) The primary rodent reservoir is the multimammate rat, which is found in Western Africa (Table 1). (17) The majority (~80 percent) of individuals infected with LASV are asymptomatic and only ~20 percent of infected individuals progress to severe disease (Table 2). (17) Lassa fever progresses through 3 stages. The first is the viral prodrome, characterized by non-specific febrile symptoms lasting 4 to 7 days. The peak phase occurs next and is characterized by sore throat, chest pain, conjunctival injection, vomiting, diarrhea, and abdominal pain. These symptoms may then progress further to hypotension, shock, neurological symptoms (e.g., altered consciousness, seizures, tremors, hearing loss), head and neck edema, and hemorrhage from the mouth, nose, rectum, or vagina (Table 2). Overt bleeding can occur in up to 40% of cases and has a strong association with mortality (Table 2). (17) The final phase is the convalescent phase. Survivors may suffer from long-term complications, such as hearing loss, seizures, cognitive impairment, and ophthalmic complications. (45)

Phenuiviridae: Rift Valley fever virus

RVFV belongs to the genus *Phlebovirus*, which has 67 species, but only 11 of those species, including RVFV, cause disease in humans. (46) RVFV geographic distribution follows that of its primary reservoir and vector, *Aedes* mosquitoes, and other mosquito and arthropod species that serve as secondary vectors (Table 1)(Figure 2). (19) While mosquitos can infect humans after taking a blood meal, most people become infected through contact with

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tissues and fluids from infected domesticated livestock. (19) RVFV induces severe and lethal disease in domesticated ruminates, causing large economic impacts in affected communities. Most humans infected with RVFV experience mild or subclinical symptoms. (19, 21) Due to the mild nature of many RVFV infections in humans and differences between outbreaks, the exact incidence of RVFV is unclear. However, estimates suggest that half a million people were infected with RVFV between 1997 and 2010. (18) Less than 1 percent of infected individuals develop hemorrhagic manifestations (Table 2). (19, 21) Initial symptoms begin in a manner similar to other VHFs, with non-specific febrile symptoms for 3–4 days. Hemorrhagic disease may begin about 1 week after symptom onset with the development of a macular rash, extended ecchymoses, gum or gastrointestinal bleeding, and bloody stools or vomit. (21) Hemorrhagic symptoms frequently portend a fatal outcome. While the overall CFR for RVFV is low (less than 1 percent), it can be up to 50 percent in cases with hemorrhagic symptoms (Table 2). Other severe disease manifestations for RVFV infection include ocular (up to 10 percent of cases) and encephalitic (less than 1 percent of cases) symptoms. (19, 21)

Nairoviridae: Crimean-Congo hemorrhagic fever virus

The *Hyalomma* tick is the primary reservoir and vector of CCHFV (Table 1)(Figure 2). Humans, livestock, and wild animals can become infected with CCHFV through the bite of an infected tick. Humans may also become infected upon handling parts or fluids of infected animals. Most infections (up to 88 percent) with CCHFV are thought to be asymptomatic. (33) Of the approximately 12 percent of infections that are severe enough to cause individuals to seek medical help, hemorrhagic symptoms (e.g., nosebleeds, ecchymoses, and blood vomit) can occur in up to 50 percent of individuals (<6 percent of all infections)(Table 2). (33) As with other VHFs, initial symptoms of CCHFV infection are non-specific and febrile. (6) A week or less after symptom onset, the hemorrhagic phase begins. Hemorrhagic disease is characterized by mucous membrane and skin petechia and extended ecchymoses, nosebleeds, coughing up blood (hemoptysis), bleeding from injection sites, and bloody stools, vomit, or urine (Table 2). These symptoms usually persist for 2–3 days. Severe cases may develop disseminated intravascular coagulation (DIC), hypovolemic shock, or multi-organ failure, including that of the liver and lungs. Survivors can suffer from a variety of complications, including hypotension, arrhythmias, difficulty breathing, and issues with their vision, hearing, or memory. Long-term complications remain undefined. (6)

Hantaviridae: Hantaan virus (HTNV) and sin nombre virus (SNV)

Orthohantaviruses are divided into Old World (e.g., HTNV) and New World (e.g., SNV) based on their geographic distributions. Old World viruses are primarily found throughout Europe and Asia, while New World viruses are found primarily in the Americas (Table 1)(Figure 2). (20) Of the 60 different viruses in the genus *Orthohantavirus*, 28 viruses, including HTNV and SNV, are clinically important. (7, 20)

HTNV and other Old World *Orthohantaviruses* cause hemorrhagic fever with renal syndrome (HFRS). Moderate (50 percent) and severe disease (20 percent) account for up to 70 percent of clinical HTNV cases and are differentiated by severity of symptoms such as bleeding (Table 2). (34) The overall CFR for HTNV infection is about 1 percent. (7, 20)

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HFRS can be divided into 5 phases. (7, 34) The first is a non-specific febrile phase lasting a week or less that progresses into a hypotensive phase. Hemorrhagic manifestations occur in less than 30 percent of cases and may begin near the end of the febrile phase. (34, 39) Hemorrhagic symptoms include conjunctival hemorrhages and petechiae in the mouth (Table 2). (7) The hypotensive phase is variable in duration (hours to a couple of days) and is characterized by hypotension that may lead to hypovolemic shock. Hemorrhagic signs may increase during this phase, including bruising, petechiae on the skin and mucosal surfaces, reddening of the conjunctiva, nose bleeds, intracranial hemorrhage, and bloody vomit, urine, or stools (Table 2). (7) Following this phase is the oliguric phase, which lasts less than a week and is characterized by temporarily decreased kidney function, which leads to low urine output (oliguria) or other abnormal characteristics (e.g., no urine output, increased protein or blood in the urine). Up to a third of individuals with severe HFRS die during the hypotensive phase and about 50 percent die during the oliguric phase. (7) The next phase is the polyuric phase, where the body begins to return to normal, including a recovery of kidney function, including urinary output. This phase can last up to several weeks. The final phase is convalescence, which can last up to half a year. Chronic renal failure and hypotension are rare complications of HFRS. (7)

SNV and other New World *Orthohantaviruses* cause hantavirus cardiopulmonary syndrome (HCPS). Like HFRS and other VHFs, HCPS begins with a non-specific febrile phase that is usually short (< 1 week). (7) The cardiopulmonary phase onsets suddenly in more than 50 percent of cases with the rapid development of a cough, shortness of breath, non-cardiac pulmonary edema, pleural effusions, hypotension, and tachycardia. Cardiogenic shock, lactic acidosis, respiratory failure, and decreased plasma volume may also occur in severe cases. (7) HCPS has a high CFR relative to HFRS (35 percent vs 1 percent)(Table 2). Individuals surviving the cardiopulmonary phase enter the polyuric phase, where pulmonary edema resolves and the body begins to return to normal. Convalescence is the final stage. While recovery is slow, survivors do not typically have any long-term complications that are directly attributed to viral infection rather than intense hospitalization. Notably, while SNV is considered a HFV, severe bleeding is not a hallmark symptom of the pathology (Table 2). (7, 20)

Flaviviridae: Yellow fever virus and dengue virus

YFV is an *Orthoflavivirus*. There are 53 total *Orthoflavivirus*es, with over half causing disease in humans. Yellow fever (YF) begins with non-specific febrile symptoms. Approximately 20–60 percent of individuals enter a remission phase for up to 2 days before progressing to the toxic phase, which is marked by yellowing of the skin or eyes (leading to the name YF) caused by liver and renal dysfunction or failure, thrombocytopenia, and hemorrhagic symptoms (Table 2). (9, 10) Hemorrhagic manifestations occur in approximately 12 percent of cases and include petechiae, bruising, bloody gums, and bloody vomit. Prior to death, individuals may also experience encephalitic manifestations, such as confusion, seizures, and coma. For individuals surviving YF, long-term complications are rarely reported. (9, 10)

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DENV is also an *Orthoflavivirus*. The disease caused by DENV has historically been broadly classified as an acute febrile illness [Dengue fever(DF)], a severe illness with fever, hemorrhagic symptoms, thrombocytopenia, and evidence of plasma leakage, such as increased/progressively increasing hematocrit or decreased albumin in the plasma or blood (Dengue hemorrhagic fever), or dengue hemorrhagic fever with additional circulatory failure (Dengue shock syndrome). In 2008, the World Health Organization reclassified the DENV presentations as DF, DF with warning signs, and severe DF. The primary difference between DF with warning signs and severe DF is the severity of the symptoms, where individuals with severe DF have severe plasma leakage, which can lead to Dengue shock syndrome, severe bleeding, and severe spleen and liver enlargement. (47) All types of DF can be broadly divided into the febrile, toxic, and recovery phases. Signs of severe disease occur during the toxic phase, including hemorrhagic symptoms. Hemorrhagic manifestations occur in 0.5–5 percent of clinically presenting DENV cases and include petechiae, bruising, and bloody vomit and stool (Table 2). (11, 16) There are 4 distinct DENV serotypes, which can be thought of as distinct viruses. Importantly, most severe DENV infections occur upon reinfection with a different DENV serotype. This is due to antibody-dependent enhancement, in which antibodies from the first infection with one DENV serotype bind and enhance infection of a different DENV serotype by facilitating entry into target cells expressing Fcγ receptors. This aspect of DENV infection has also complicated vaccine development (Table 1). (43)

Section 2: The hemostatic system

Hemostasis is defined as the processes that stop bleeding following blood vessel injury. The hemostatic system has several different components, including platelets, the coagulation system, the fibrinolytic system, the endothelium, and the blood vessel wall. Under normal conditions, these components are in balance, leading to the formation of a hemostatic plug comprised of platelets and cross-linked fibrin that stops bleeding. However, during disease, the hemostatic system can become dysregulated, leading to bleeding and thrombosis.

The coagulation system

The coagulation system contains pro-coagulant factors that are required for the formation of fibrin and anti-coagulant factors that regulate different points of the coagulation cascade. The coagulation cascade can be divided into the extrinsic, intrinsic, and common pathways. The extrinsic pathway [tissue factor (TF)/Factor (F) VIIa complex] primarily initiates blood coagulation. Under normal conditions, TF is expressed by cells underlying the vascular endothelium. (48) Damage to a blood vessel leads to the activation of the coagulation protease cascade by the TF/FVIIa complex. (48) The intrinsic pathway is initiated by the activation of FXII. (49) FXIIa activates FXI to FXIa, which then activates FIX to FIXa. FIXa in association with its cofactor FVIIIa forms the intrinsic Xase complex. The common pathway includes the prothrombinase complex (FXa/FVa) that converts prothrombin to thrombin. Thrombin is the central protease of the coagulation cascade and cleaves fibrinogen to form fibrin monomers and activates the transglutaminase FXIII, the cofactors FV and FVIII, as well as platelets. (49) It should be noted that FXIIa can also activate pre-kallikrein

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to kallikrein, which cleaves high molecular weight kininogen resulting in the liberation of bradykinin, which increases vascular permeability. (49)

Fibrinogen is comprised of 2 copies each of 3 chains: Aα, Bβ, and γ. (50) Fibrinogen is assembled in hepatocytes and secreted as a soluble protein. (50) Upon cleavage by thrombin, fibrinogen is converted to fibrin monomers. These monomers oligomerize and eventually form protofibrils. These protofibrils aggregate to form an insoluble fibrin mesh. The stability of the fibrin mesh is further enhanced through cross-linking by FXIIIa. (50)

Platelets

Platelets are small, anucleate blood cells produced from megakaryocytes in the bone marrow. (51) They are rapidly recruited to sites of vessel injury. Initial platelet tethering to the injured vessel is mainly mediated by the platelet receptor GPIb-V-IX binding von Willebrand Factor (vWF) that is deposited from the plasma or released from activated endothelial cells (ECs). Firm adhesion requires the interaction of various platelet integrins with components in the extracellular matrix. Next, binding of soluble agonists, such as thrombin, ADP, or thromboxane A2, to platelet G protein-coupled receptors activates the platelets. Finally, binding of fibrinogen to activated $\alpha_{IIb}\beta_3$ on platelets mediates platelet aggregation. (51)

The endothelium

The vascular endothelium is an anti-thrombotic surface that inhibits the activation of platelets and coagulation. (52) For example, ECs inhibit platelet activation by expressing CD39, which degrades ATP, and by releasing nitric oxide and prostacyclin. The endothelium also expresses anti-coagulant proteins, such as tissue factor pathway inhibitor and components of the protein C system (e.g., thrombomodulin, endothelial protein C receptor), as well as molecules (e.g., heparin sulfates) that bind antithrombin. In addition, ECs release tissue plasminogen activator (tPA), which promotes fibrinolysis. (52) The integrity of the EC barrier is maintained by tight junctions and adherens junctions between adjacent ECs and the EC glycocalyx. (53, 54) Under pathological conditions, however, the endothelium is converted into a prothrombotic surface by the release of vWF and P-selectin from Weibel-Palade bodies, down regulation or shed of anticoagulant proteins (e.g., thrombomodulin), and induction of TF expression. (52)

The fibrinolytic system

The fibrinolytic system generates plasmin that degrades cross-linked fibrin. tPA and urokinase plasminogen activator (uPA) activate plasminogen to plasmin. (55) tPA and plasminogen bind to fibrin, which enhances tPA activation of plasminogen. uPA binds to urokinase plasminogen activator receptor on cells to activate plasminogen. Upon activation, plasmin cleaves cross-linked fibrin to generate fibrin degradation products (FDPs), such as D-dimer. (55) tPA and uPA are inhibited by plasminogen activator inhibitor (PAI)-1 and PAI-2. PAI-1 has a higher affinity for tPA, while PAI-2 has a higher affinity for uPA. Plasmin is inhibited by α2-antiplasmin. Thrombin also activates thrombin activatable fibrinolysis inhibitor, which cleaves plasminogen binding sites in fibrin. (55)

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Section 3: Mechanisms of hemostatic derangement in viral hemorrhagic fever

In this section, possible mechanisms leading to bleeding during VHF are discussed. Many different biomarkers have been measured in both humans and non-human primates (NHPs) during VHF. Unfortunately, most human data can be thought of as a single snapshot that may or may not accurately represent the total infection course. Additionally, most time points from humans are based on symptom onset. Because many of the initial symptoms of VHF can be mild and non-specific, pinpointing the exact timepoint during infection when a sample was acquired from an infected individual can be difficult and are estimates at best. Thus, data from NHP models have been key to exploring and elucidating the mechanisms underlying bleeding during VHF. Tables 3 and 4 summarize various hemostatic-related parameters and biomarkers in both humans and NHPs infected with the different HFVs, respectively.

Bleeding can arise from either consumption of coagulation factors and platelets and/or inadequate synthesis of coagulation factors and platelet production. Coagulation activation is part of the host response to viral infection and is thought to limit pathogen dissemination and prevent plasma leakage resulting from damage due to infection. (129) However, viral infection can lead to DIC that results in the consumption of coagulation factors (consumptive coagulopathy) and platelets (thrombocytopenia), leading to bleeding. The International Society on Thrombosis and Hemostasis recently updated the scoring criteria for the clinical diagnosis of DIC (Table 5). (130) An increase in the prothrombin time and a decrease in fibrinogen indicate the consumption of coagulation factors. A low platelet count indicates thrombocytopenia, whereas D-dimer is a marker of both coagulation activation and fibrinolysis. Severe coagulopathies, including DIC, are life threatening. (130)

The coagulation system is activated during viral hemorrhagic fever.

Coagulation activation is a feature common to nearly all HFV infections. The TF/FVIIa complex is the primary activator of coagulation in vivo, and TF expression is increased in many diseases. (48) TF is a transmembrane protein that localizes FVII/FVIIa to the surface of cells and extracellular vesicles (EVs). Measuring TF in plasma is difficult due to its low abundance. (131) Importantly, activity-based assays are more sensitive than antigen-based assays. (132)

A landmark study demonstrated that EBOV increases TF expression in peripheral blood mononuclear cells (PBMCs, notably the monocyte) in vitro and in vivo using a NHP model. (98) Whether other cell types also upregulate TF following EBOV infection is unknown. Additionally, a recent study showed that there is increased EV-TF activity in plasma from EBOV-infected NHPs. (128) Further, TF expression is central to EBOV pathogenesis, as administration of the TF/FVIIa inhibitor NapC2 reduced disease (e.g., delayed appearance of characteristic rash and lower D-dimer) and prolonged and increased survival in EBOV-infected NHPs. (119)

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Infections with MARV in a NHP model also led to increased TF expression in PBMCs compared to controls. (121) However, the administration of NapC2 was not protective in a MARV NHP model. (121) The authors noted that the virus isolate used was more virulent than other strains previously used in the same NHP model. (121) While inhibition of the TF/FVIIa complex was not effective in preventing severe disease in this MARV model, it does not preclude the possibility that TF/FVIIa inhibition may be effective for less virulent MARV strains.

Several studies have measured plasma levels of TF antigen in DENV-infected individuals. It is difficult to compare these studies because they use different assays to measure TF and samples from patients with different disease severities. One study found increased plasma TF in severe DF compared to DF. (101) Interestingly, the healthy control group and the DF group did not differ significantly. This could be due to the timing of sample collection, as samples were collected 1–15 days post symptom onset, which on the extreme end may be past the window of induction. (101) A study examining TF expression in the plasma of children with severe DF at different times during disease observed children with severe DF had high levels of plasma TF at hospital admission that decreased over time. (99) Additionally, PBMCs from individuals with severe DF had increased TF protein expression compared to those with DF or healthy controls. (100) A fourth study found a trend towards increased EV-TF activity in individuals with DF compared to healthy controls. (133) However, the point during disease that the samples were collected is unclear. (133)

Individuals with LASV had higher levels of plasma TF protein compared to non-LASV febrile controls. (80) In fatal cases of RVFV, plasma TF antigen is also increased. (88) An increase in plasma EV-TF activity was also observed in individuals infected with SNV. (81) However, the cellular source of TF during LASV, RVFV, and SNV infections remains to be explored. An increase in EV-TF activity in the plasma of individuals infected with Puumala virus, a close relative of HTNV, has also been noted and associated with intravascular coagulation. (134)

TF levels during other HFV infections have not been measured. Additionally, because all the HFVs examined in this review are enveloped, the new virions derived from TF-positive host cells may incorporate TF into the viral envelope as has been shown for several herpesviruses. (135) However, this remains to be established for HFVs. In summary, increased TF on PBMCs (likely monocytes), other cell types, and EVs is likely the primary activator of the coagulation cascade observed in VHF (Figure 3). Understanding the triggers that activate coagulation in VHF may identify new therapeutic targets to prevent coagulation activation during VHF.

Viral hemorrhagic fever is associated with liver damage.

VHF is associated with liver damage in humans and NHP models, as evidenced by increased aspartate aminotransferase and alanine aminotransferase and liver pathology. (58, 61–63, 67–70, 74, 77, 78, 80, 88, 91–95, 97, 115, 136–144) The liver is the primary site for synthesis of most coagulation factors, including prothrombin and fibrinogen. Hepatocytes are also the primary source of the anti-coagulant protein C. Indeed, the level of protein C decreased in humans and NHPs infected with HFVs (EBOV, MARV, LASV, YFV, and

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DENV). (2, 70, 73, 98, 109, 111, 119, 121, 122) Several of the HFVs (EBOV, MARV, LASV, RVFV, CCHFV, and YFV) are hepatotropic, meaning they directly infect hepatocytes (Table 1). Some of these viruses are also cytopathic in cultured hepatocytes (EBOV, LASV, CCHFV, YFV) while others are not (MARV). (145–149) From examination of pathological specimens, hepatocytes stain positive for viral antigen and show signs of apoptosis and necrosis. However, whether this is due to viral replication or immune-mediated mechanisms cannot be determined. In addition to directly killing hepatocytes, these viruses may also lead to generalized hepatocyte dysfunction upon infection, which is further increased by severe inflammation. For the non-hepatotropic viruses (HTNV, SNV, and DENV), general liver dysfunction or pathology likely arises through immune-mediated mechanisms. Ultimately, both processes lead to a reduction in coagulation factor and protein C synthesis (Figure 3).

Thrombocytopenia and platelet dysfunction during viral hemorrhagic fever.

Thrombocytopenia can be due to decreased platelet production, increased platelet activation and consumption, or a combination thereof. The role of platelets during HFV infection is best characterized for DENV. (150) Both decreased platelet production and increased activation and consumption contribute to thrombocytopenia during DENV infection. Transient bone marrow suppression and infection of megakaryocytes may contribute to decreased platelet production early during infection. (151–153)

During the toxic phase of DENV infection in which thrombocytopenia is a key feature, the bone marrow is normal, suggesting increased platelet clearance and/or consumption is underlying thrombocytopenia. Platelets isolated from individuals infected with DENV are activated, exhibiting increased surface expression of active $\alpha_{IIb}\beta_3$, P-selectin, and phosphatidylserine (PS). (154–157) Platelets from individuals with DENV are also less responsive to in vitro agonist stimulation compared to those from convalescent individuals or healthy controls. This suggests that platelets from infected individuals may be exhausted and have reduced function, and as such they cannot adequately respond to stimuli. (156) Additionally, some platelets isolated from individuals with DENV are apoptotic. (154, 155)

Platelet activation likely occurs through several mechanisms during DENV infection. Both viral RNA and antigen have been detected in isolated human platelets. (157, 158) This observation in combination with human platelets supporting DENV replication in vitro suggests that DENV may replicate to some extent in platelets in vivo. (157–159) However, whether platelets produce new virions in vitro or in vivo is hotly debated and platelets likely do not contribute to viremia. (160) The viral protein NS1, which is released into the circulation during infection, can also activate platelets directly. (161, 162) The levels of NS1 are positively correlated to disease severity and the development of thrombocytopenia. (162) Indeed, in vitro NS1-stimulated platelets have increased levels of PS and P-selectin surface expression, as well as pro-longed aggregation following in vitro agonist stimulation. (163) In a DENV mouse model that develops thrombocytopenia, genetic deletion of NS1 prevented thrombocytopenia. (163). Activated platelets may also bind to activated ECs. (163) EC-bound platelets are removed from the circulation and cannot be measured during routine blood counts, ultimately contributing to the observed thrombocytopenia.

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Activated platelets express various receptors that mediate their clearance. (164) Both increased PS exposure and the coating of platelets with antibodies or complement enhance engulfment by phagocytes. Both anti-platelet antibodies and antibody- and complement-coated platelets have been isolated from individuals with DENV. (157, 165) Moreover, increased platelet-phagocyte aggregates have been isolated from individuals with DENV; additionally, phagocytosis of platelets isolated from individuals with DENV compared to healthy controls was increased in vitro and this correlated with disease severity. (154) Exposure of platelets to NS1 also led to increased phagocytosis in vitro, suggesting a mechanism by which phagocytosis may be stimulated in vivo. (163) Platelets may also be removed by phagocytes through opsonization or lysed via complement activation and subsequent formation of the membrane attack complex. (165, 166)

Platelets can also contribute to inflammation during DENV infection via the secretion of various cytokines, which can activate other cell types and increase vascular permeability. (150) DENV-infected platelets may also be an important source of secreted NS1, which directly contributes to increased vascular permeability in addition to its role in platelet activation. (167)

Thrombocytopenia occurs mainly during severe LASV infection and rarely in non-severe infection. In NHPs, platelet survival time is normal. However, platelets from both humans and NHPs with severe LASV infection exhibit aggregation defects in response to in vitro agonist stimulation. (80, 110, 168) In a follow up study, human plasma from severe LASV cases inhibited aggregation of platelets collected from healthy donors. Platelet aggregation activity was restored following washing of the platelets, suggesting the presence of a platelet inhibitor in the plasma of individuals with severe LASV. (169) Aggregation curves also show that platelets from individuals with LASV disaggregate, which may be suggestive of defects in granule release, which is known to sustain platelet aggregation over time. (80, 168) Improved aggregation responses also coincided with clinical improvement, suggesting that platelet inhibition contributes to the pathology of LASV. (168)

Pathogenic *Orthohantaviruses* (e.g., HTNV and SNV) bind the inactive form of β_3 integrins. (170) This may lead to defective platelet aggregation and adhesion responses. Indeed, regardless of illness severity, platelets from individuals with HFRS exhibit aggregation and granule release defects in response to in vitro agonist stimulation that improve as illness resolves. (171) However, the defects in severely ill individuals were worse compared to moderately and mildly ill individuals. These properties appear to be intrinsic to the platelets themselves, rather than due to their environment (e.g., plasma), as mixing normal platelets with HFRS plasma led to normal aggregation. Additionally, quiescent platelets bind infected ECs via cell-surface displayed viral glycoproteins in vitro, which may lead to reduced circulating platelet numbers. (170)

Individuals with CCHFV have an increased immature platelet fraction and mean platelet volume compared to healthy individuals. (172) This suggests that the bone marrow is likely not suppressed and that thrombocytopenia in CCHFV is due to increased activation and consumption rather than decreased production. However, studies on bone marrow specimens show conflicting results, with some showing a normal marrow with

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respect to megakaryocyte numbers and platelet production and others showing decreased megakaryocyte numbers. (66, 173, 174) It remains unclear if platelets are activated or if they exhibit defective aggregation responses during CCHFV infection.

The state of platelets during EBOV and MARV infections is unclear. Excessive coagulation activation was thought to be the primary driver of thrombocytopenia during EVD and MVD for many years. This was because very little data exists on platelet function during EVD and MVD in humans or NHPs. One study using an NHP model of EBOV infection showed platelet aggregation defects in response to in vitro agonist stimulation that coincided with increased levels of plasma platelet factor-4 in vivo. (117) These results suggest increased platelet activation and degranulation during infection. Effects of EBOV on the bone marrow are conflicting, with a singular report from an unspecified number of patients suggesting normal bone marrow cellularity, including megakaryocyte number; in contrast, a more recent and thorough investigation on the bone marrow from NHPs infected with EBOV suggests EBOV can infect megakaryocytes and induce bone marrow suppression. (136, 175)

Mechanisms underlying thrombocytopenia in YFV and RVFV infections remain to be elucidated.

Infection with the hemorrhagic fever viruses induces endothelial cell activation and increased vascular permeability.

HFV infection leads to EC activation, which is broadly characterized by a loss in anti-thrombotic properties, expression of pro-coagulant, pro-adhesive, and pro-inflammatory molecules, and increased vascular permeability. This can occur directly through viral infection, by the action of specific viral proteins, or through inflammatory mediators released in response to viral infection. These mechanisms likely co-occur for many of the HFVs and can lead to disrupted barrier function (i.e., the loosening of junctions between cells), increased vascular permeability (the loss of EC barrier function in vivo), and increased vascular leak (the leakage of cells and proteins from the blood into the tissue).

Regardless of whether ECs are direct targets or sustain any outward damage, infection with HFVs leads to EC activation and disruption, marked by increases in the circulation of 1 or more of the following proteins: vWF, soluble (s)-ICAM-1, s-VCAM-1, and s-P-selectin. Viral antigens from most of the HFVs have also been found in ECs from different tissues, and all of them can replicate in cultured ECs. However, only some of the viruses have evidence of replication in ECs in vivo (e.g., EBOV, MARV, CCHFV, HTNV, SNV, YFV, DENV) and even fewer (e.g., MARV, RVFV, and DENV) are directly cytopathic in ECs in vitro. (8, 176–179) This suggests there are alternative mechanisms leading to the observed increases in vascular permeability during VHF.

How each virus alters ECs ultimately shapes the observed pathology. For example, severe HTNV infection manifests as HFRS, and concordantly the strongest viral antigen staining is observed in the kidney, particularly renal capillary ECs. (180) In contrast, SNV infection presents as HCPS, and the ECs lining the lung capillaries and other small vessels stain the strongest for viral antigen. (8) Indeed, pathogenic *Orthohantaviruses* utilize β_3 integrins in the inactive conformation as entry receptors, including $\alpha_{IIb}\beta_3$ and $\alpha_V\beta_3$, expressed

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by endothelial cells and platelets. (181) Importantly, neither HTNV nor SNV kill ECs directly but drastically increase vascular permeability. (8, 180) Markers of EC activation (s-E-selectin, s-ICAM-1, s-VCAM-1, and VEGF) and a marker of degradation of the EC glycocalyx (syndecan-1) are elevated in the circulation in individuals with HFRS or HCPS (Table 3). (106, 182) In vitro work suggested several possible mechanisms by which HTNV and SNV may increase vascular permeability. The first mechanism suggested that virus binds to the surface of the endothelium and inhibits binding of host proteins to β_3 integrins. β_3 integrin inhibition then leads to increased VEGF-induced vascular permeability in ECs. (183, 184) The second mechanism builds on of the first and posits that increased VEGF combined with hantaviral inhibition of $\alpha_V\beta_3$ integrin, leads to internalization of VE-cadherin and a redistribution of the junctional protein ZO-1, resulting in decreased barrier function. (15, 184–189) However, a later study using a more sophisticated in vitro capillary system did not observe changes in VE-cadherin or VEGF levels. (190) Rather, in the third proposed mechanism, increased permeability was the result of increased FXII binding and autoactivation and increased kallikrein, which together led to increased bradykinin and increased vascular permeability. (190) Whether 1 or any of these mechanisms occur in vivo remains unclear. However, all ultimately would lead to loosened EC junctions, resulting in decreased barrier function in vitro and increased vascular permeability in vivo without cell death.

For other viruses, such as EBOV, and likely MARV, ECs are targets late in infection but appear to have minimal damage in vivo. (136, 191) The viral glycoprotein GP is known to induce EC activation and changes in EC barrier function. (192–195) These changes are further enhanced by the presence of TNFα, which is present in the blood at high levels during EBOV and MARV infections. (2, 196, 197) There are increased levels of various markers of EC activation in the circulation, including s-thrombomodulin, s-P-selectin, s-PECAM-1, s-ICAM-1, with some being associated with severe disease (s-thrombomodulin, s-P-selectin, s-PECAM-1, s-ICAM-1), hemorrhage (s-ICAM-1, s-thrombomodulin) or death (s-thrombomodulin) (Table 3). (84, 85)

There is evidence for multiple mechanisms of decreased EC barrier function and increased vascular permeability and leak in DENV and YFV infections. DENV NS1, a viral protein, induces decreased EC barrier function directly. (198) While DENV NS1 reduced the barrier function of ECs of various origins in vitro, YFV NS1 decreased the barrier function of only liver ECs. (199) Data suggests that both DENV and YFV NS1 also induce glycocalyx disruption in vitro. (198, 199) Indeed, sydecan-1 is elevated in individuals with severe DENV and YFV infection (Table 3). (107, 108) There is also evidence that DENV may replicate in ECs. However, similar evidence for YFV is lacking. (176) Flavivirus NS2 and NS4 have also been suggested to have viroporin-like activities, which may contribute further to increased permeability, but this remains to be established. (200) Both DENV and YFV infections are associated with increased levels of various cytokines, which can further enhance the vascular permeability induced by viral proteins and viral replication. (201, 202)

For CCHFV and LASV, there is histopathological evidence of outward EC damage, but it remains unclear if this is virus- or immune-mediated. (13, 203) CCHFV activates ECs via infection in vitro in a dose-dependent manner, and the viral protein GP38 decreased EC

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barrier function in vitro by disrupting the EC glycocalyx. (204, 205) GP38 also increased vascular permeability and leak in vivo in a CCHFV mouse model. (205) During CCHFV infection plasma levels of vWF and s-ICAM-1 are significantly elevated compared to healthy controls (Table 3). (104) In a separate cohort, s-ICAM-1, s-VCAM-1, and VEGF were higher in individuals that died from CCHFV infection compared to those who survived (Table 3). (104, 105) LASV infection is also associated with increases in plasma EC activation markers (e.g., vWF, s-ICAM-1, and s-VCAM-1) in the plasma compared to non-LASV febrile controls or healthy controls (Table 3). (80) Additionally, in vitro data suggests LASV is not cytopathic to ECs. (80, 206)

Markers of EC activation (VEGF, s-PECAM-1, s-E-selectin) are moderately elevated in the circulation of individuals infected with RVFV compared to healthy individuals (Table 3). (87) In RVFV infection, no staining for viral antigen is observed in ECs from infected domesticated ruminant or mouse tissues, which suggests an immune-mediated mechanism of EC activation. (207, 208)

While increased vascular permeability and leak contribute to the shock and multi-organ failure observed in end-stage VHF, increased vascular permeability also causes cells underlying the vascular endothelium to become exposed to the circulation. Importantly, some of these cells are a source of high levels of TF. When exposed to the circulation at the scale occurring during VHF, this likely contributes to coagulation activation. In addition, activation of ECs leads to down regulation of anti-coagulant proteins, exacerbating the hypercoagulable state.

Phosphatidyl serine exposure and viral mimicry

PS is normally located on the inner leaflet of the cell membrane. During damage or cellular death, PS is flipped to the outer membrane. Normally, this serves as an “eat me” signal to phagocytes and results in clearance of the damaged cells. (209) In addition to dying cells, activated platelets also have increased PS exposure. (209) PS also contributes to coagulation as PS binding to TF leads to a conformational change in TF that enhances the pro-coagulant activity of the TF/FVIIa complex. (210) Additionally, PS provides a surface for the assembly and function of coagulation proteases/cofactor complexes. (48)

During viral infection, both dying cells and activated platelets are important sources for PS. Additionally, PS may be incorporated into both the viral envelope of new virions and into EVs released from PS-positive cells. Indeed, EBOV, MARV, LASV, YFV, and DENV have all been shown to incorporate PS into their viral envelopes, facilitating viral entry into host cells using receptors that bind PS (e.g., T cell immunoglobulin and mucin receptor family protein and TYRO3–AXL–MER (TAM) family proteins). (211) This mechanism of entry not only increases the ability of the virus to enter phagocytes, a preferred target of infection for many HFVs, but may also promote coagulation. However, coagulation activation directly by viral particles remains to be established for HFVs as has been done for other viruses (Cytomegalovirus and Herpes Simplex Virus-1 and −2). (135, 212).

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Fibrinolysis during viral hemorrhagic fever

Dysregulation of the fibrinolytic system may contribute to bleeding in VHF. Indeed, D-dimer is elevated to some extent in all VHFs considered herein. Additionally, tPA is elevated in humans or NHPs infected with EBOV, LASV, RVFV, SNV, and DENV, and levels of plasmin anti-plasmin complexes are elevated in individuals with severe DF (Tables 3–4)(2, 15, 62, 63, 66, 67, 70–74, 79, 81–91, 98, 102, 110–112, 117, 119–122) (73) However, PAI-1 is also increased in humans with the same viruses, which may limit the activation of the fibrinolytic system by tPA (Table 3). (72, 73, 80, 84, 86, 87, 102, 103)

To better understand the kinetics of when these elevations occur, animal models are essential. In a NHP model of EBOV infection, D-dimer begins to rise at 1-day post-infection and continues to increase over the course of infection. Both tPA and uPA increased by day 4 post-infection. (98) Unfortunately, PAI-1 could not be measured in this model because the available assays were incompatible with NHP samples. (98) In 2 other studies using transcriptomics and a multiplex bead-based assay on whole blood from NHPs infected with a different EBOV strain, PAI-1 and tPA both increased. (123, 213)

Very few studies have examined biomarkers of fibrinolysis other than D-dimer or total PAI-1. While these biomarkers can suggest increased fibrinolysis, other biomarkers such as plasmin anti-plasmin complexes or tPA/PAI-1 complexes, may be better metrics of fibrinolytic activity. As such, it is presently unknown whether fibrinolytic dysregulation contributes directly to bleeding in VHF. Thus, more studies on the fibrinolytic system in both humans and NHPs are needed.

Summary and perspectives

VHF is a catch-all term, encompassing several different diseases caused by viruses from distinct families. Bleeding is the symptom for which the term VHF was coined, but bleeding is not necessarily a feature in all cases. Regardless, bleeding is a severe manifestation of the spectrum of diseases caused by HFVs. Herein, we propose some potential common mechanisms underlying bleeding in VHF (Figure 3). Viral infection activates coagulation via inducing increased TF expression on infected cells. The activity of the TF/FVIIa complex is further enhanced by increased PS exposure, resulting in even more coagulation activation. It is possible that the intrinsic pathway also contributes to the activation of coagulation in VHF. In parallel, there is decreased clotting factor synthesis by the liver due to damage during infection. Activation of platelets and decreased platelet production lead to thrombocytopenia. In addition, platelet dysfunction occurs in VHF. Viral infection also activates ECs, causing decreased activity of the activated protein C anti-coagulant pathway, conversion to a prothrombotic surface, and increased vascular permeability. EC activation also leads to increased tPA release, which ultimately results in increased plasmin activity and enhanced fibrinolysis. We propose that a combination of these pathways leads to bleeding during VHF.

The pathways highlighted above are based on data primarily from individual viruses, as a complete picture for any singular virus is missing. Thus, it still remains unclear if the mechanisms discussed herein extend to all the HFVs. Some pathways, such as fibrinolysis,

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also remain largely unexplored. More studies are needed to better understand the complete picture of the hemostatic dysregulation during VHF caused by different viruses. While human biomarker data has been helpful in opening avenues of investigation, human data alone at a singular time point after an undefined amount of time after infection cannot provide the full picture. In vitro data also cannot capture the complex interplay of the coagulation and immune systems. Thus, animal models have been and will continue to be essential in advancing our understanding of mechanisms underlying bleeding in VHF. A better understanding may open new therapeutic avenues to treat and prevent bleeding during VHF.

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Non-standard Abbreviations and Acronyms

CCHFV	Crimean Congo hemorrhagic fever virus
CFR	Case fatality rate
DIC	Disseminated intravascular coagulation
EBOV	Ebola virus
EC	Endothelial cell
EV	Extracellular vesicle
EVD	Ebola virus disease
F	Factor
FDP	Fibrin degradation product
HCPS	Hantavirus cardiopulmonary syndrome
HFRS	Hemorrhagic fever with renal syndrome
HFV	hemorrhagic fever virus
HTNV	Hantaan virus
LASV	Lassa fever virus
MARV	Marburg virus
MVD	Marburg virus disease
NHP	Non-human primate
PBMC	Peripheral blood mononuclear cell

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PS	Phosphatidylserine
RVFV	Rift Valley fever virus
SNV	Sin nombre virus
TF	Tissue factor
tPA	Tissue plasminogen activator
uPA	Urokinase plasminogen activator
VHF	viral hemorrhagic fever
vWF	von Willebrand factor
YF	Yellow fever
YFV	Yellow fever virus

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Highlights

- Viral hemorrhagic fever is a term collective referring to several different diseases caused by various viruses.
- Both direct effects from infection and the host immune response contribute to hemostatic dysregulation.
- We discuss herein the potential mechanisms underlying bleeding in viral hemorrhagic fever, including increased tissue factor expression, thrombocytopenia, platelet dysfunction, decreased coagulation factor synthesis, and endothelial cell activation and damage.

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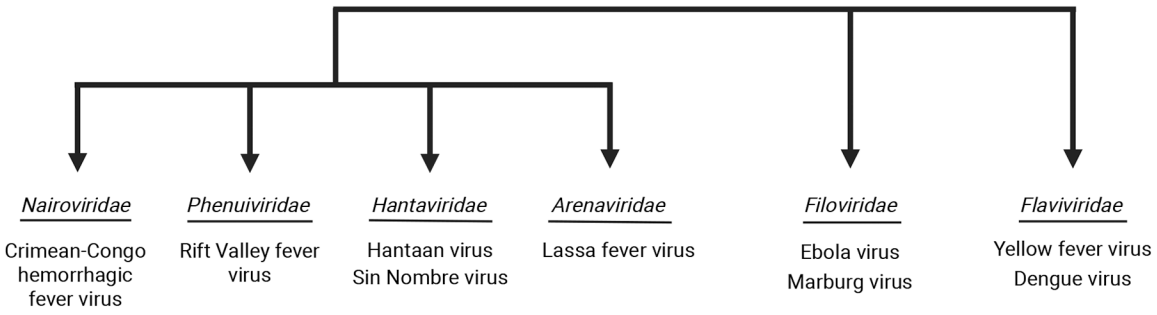


Figure 1. Overview of the phylogenetic relationship between viruses causing viral hemorrhagic fever.

An abbreviated phylogenetic relationship of the different viruses causing viral hemorrhagic fever is shown by virus family with each individual species shown below. Figure made with BioRender.com.

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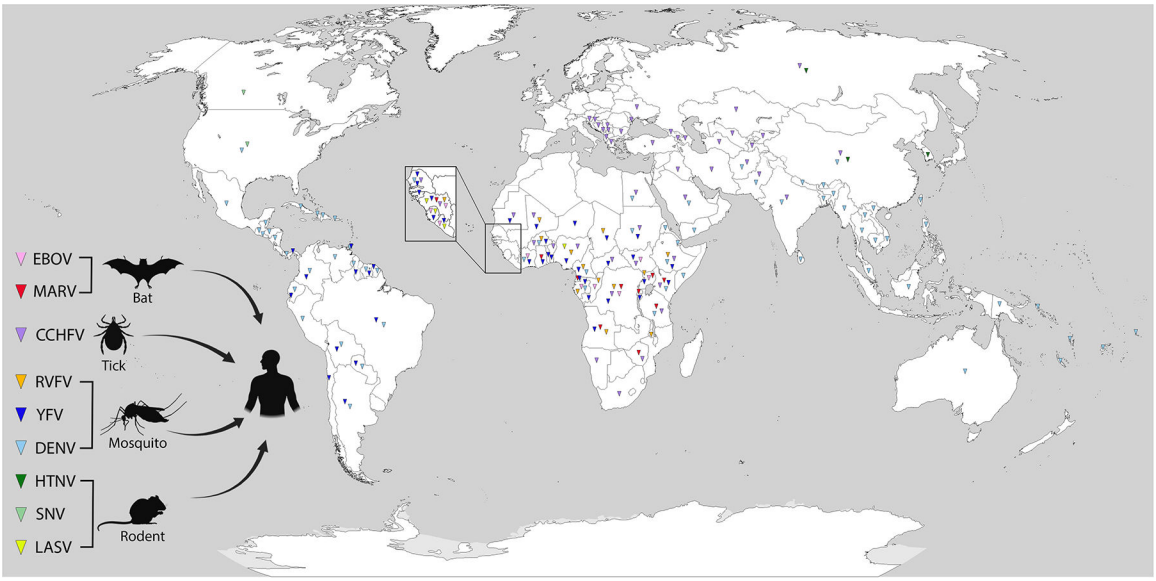


Figure 2. Geographic distribution of the different hemorrhagic fever viruses.

Inverted triangles colored coded to represent each individual virus are shown on the map where the virus is known to circulate: Ebola virus (EBOV, pink), Marburg virus (MARV, red), Crimean Congo hemorrhagic fever virus (CCHFV, purple), Rift Valley fever virus (RVFV, orange), yellow fever virus (YFV, dark blue), dengue virus (DENV, light blue), Hantaan virus (HTNV, dark green), sin nombre virus (SNV, light green), and (LASV, yellow green). The vectors and/or reservoirs for each virus are also shown, as the distribution of the viruses follows that of their vector and/or reservoir.

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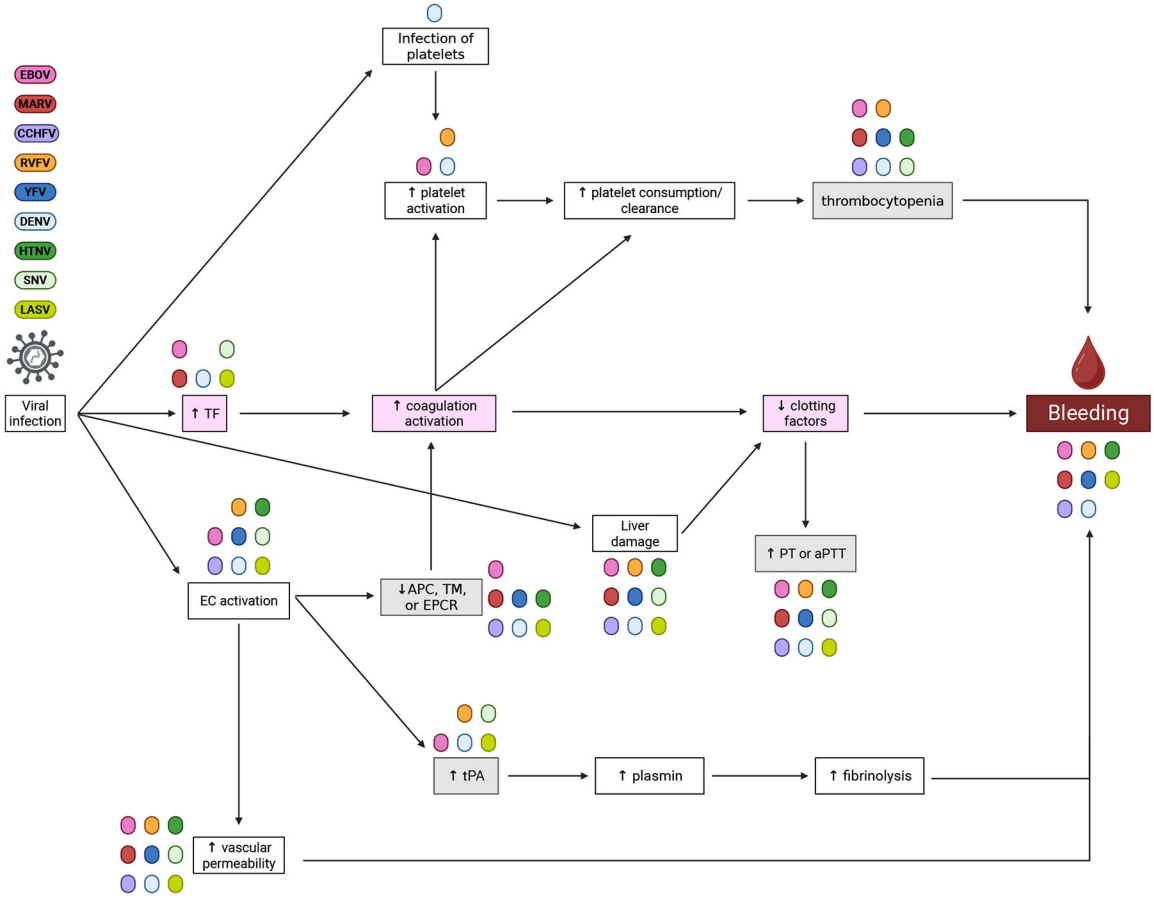


Figure 3. Potential mechanisms underlying bleeding during viral hemorrhagic fever.

Viral infection induces tissue factor (TF) expression in host cells, resulting in coagulation activation and consumption of clotting factors. Liver infection leads to decreased production of clotting factors. Coagulation activation and viral infection lead to platelet activation and clearance, resulting in thrombocytopenia. Viral infection also activates endothelial cells (ECs), leading to increased vascular permeability and decreased protein C anti-coagulant activity. Activated ECs also release tissue plasminogen activator (tPA), resulting in increased plasmin and enhanced fibrinolysis. Cumulatively, these mechanisms result in bleeding during viral hemorrhagic fever. For each potential mechanism, a colored dot corresponding to each virus that has evidence supportive of that mechanism is shown: Ebola virus (EBOV, pink), Marburg virus (MARV, red), Crimean Congo hemorrhagic fever virus (CCHFV, purple), Rift Valley fever virus (RVFV, orange), yellow fever virus (YFV, dark blue), dengue virus (DENV, light blue), Hantaan virus (HTNV, dark green), sin nombre virus (SNV, light green), and (LASV, yellow green). APC – activated protein C; TM – thrombomodulin; EPCR – endothelial protein C receptor; PT – prothrombin time; aPTT – activated partial thromboplastin time. Figure made with BioRender.com.

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Table 1.

Information on the viruses that cause viral hemorrhagic fever.

	EBOV	MARV	LASV	RVFV	CCHFV	HTNV	SNV	YFV	DENV
Virus family, genus	*Filoviridae, Ebolavirus*	*Filoviridae, Marburgvirus*	*Arenaviridae, Mammarenavirus*	*Phenuiviridae, Phlebovirus*	*Nairoviridae, Orthonairovirus*	*Hantaviridae, Orthohantavirus*	*Hantaviridae, Orthohantavirus*	*Flaviviridae, Orthoflavivirus*	*Flaviviridae, Orthoflavivirus*
Primary cellular targets of the virus	Myeloid (1)	Myeloid (2, 3)	Myeloid (4)	Hepatocytes Neurons Myeloid (5)	Hepatocytes Endothelial (6)	Endothelial Myeloid (7)	Endothelial Myeloid (7, 8)	Hepatocytes Myeloid Endothelial (9, 10)	Myeloid (11)
Organ(s) primarily affected	Liver, spleen, secondary lymphoid organs, others later in infection (1)	Liver, spleen, secondary lymphoid organs, others later in infection (3)	Liver, spleen (12, 13)	Liver (14)	Liver (6)	Kidney (7)	Lung (15)	Liver (10)	Liver and spleen (16)
Geographic distribution	Sub-Saharan Africa (1)	Sub-Saharan Africa (3)	Western Africa (17)	Africa, Saudi Arabia, Yemen (18, 19)	Africa, Middle East, Southeast Asia, Southern/ Easter Europe (6)	Asia (7, 20)	North America (7, 20)	Sub-Saharan Africa, South America (10)	Sub-Saharan Africa, Central and South America, Southeast Asia (16)
Mechanism of transmission	Contact with infected bodily fluids/ tissues (1)	Contact with infected bodily fluids/tissues (3)	Contact with infected rodent/ rodent secretions excretions (e.g., urine, feces) Contact with infected bodily fluids (17)	Mosquito bite, contact with infected bodily fluids/tissues (19, 21)	Tick bite Contact with infected bodily fluids (6)	Contact with infected rodent/ rodent secretions excretions (e.g., urine, feces) (7, 20)	Contact with infected rodent/ rodent secretions excretions (e.g., urine, feces) (7, 20)	Mosquito bite (10)	Mosquito bite (16)
Vector	N/A	N/A	N/A	*Culex* and *Aedes* spp. (19, 21)	*Hyalomma* spp. (6)	*Apodemus agrarius* (7, 20)	*Peromyscus maniculatus* (7, 20)	*Aedes* spp. Haemogogus spp. Sabethes spp. (10)	*Aedes* spp. (16) (22)
Reservoir host	Unknown – bats suspected (1)	*Rousettus aegyptiacus* (23)	Multimammate rat (17)	*Aedes* spp., rodents, wild ruminants, bats (19)	*Hyalomma tick* (6)	*Apodemus agrarius* (7, 20)	*Peromyscus maniculatus* (7, 20)	Non-human primates (10)	Humans (22)

Abbreviations: EBOV – Ebola virus; MARV – Marburg virus; LASV – Lassa fever virus; RVFV – Rift Valley fever virus; CCHFV – Crimean Congo hemorrhagic fever virus; HTNV – Hantaan virus; SNV – sin nombre virus; YFV – yellow fever virus; DENV – dengue virus

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Table 2.

Case presentation of viral hemorrhagic fever caused by different viruses.

	EBOV	MARV	LASV	RVFV	CCHFV	HTNV	SNV	YFV	DENV
Global incidence (cases/year)	Variable – usually small <300 * (3, 24)	Variable – usually small <300 (3, 25)	Estimated 100,000–300,000 (17)	Variable and outbreak dependent (18)	Estimated 10,000–15,000 ** (26)	Estimated >10,000 (20, 27)	30 (20)	200,000 (10)	>100,000,000 ** (11)
Case fatality rate (%)	60 (24)	50 (28)	1 (29)	<1 (30)	5–40 (31)	~1 (7, 20)	35 (20)	>20 (10)	<1 (11)
Cases having severe manifestations (%)	60–85 (32)	-	20 (17, 29)	<5 (30)	12 (33)	~20 (34)	>50 (35)	12 (36)	0.5–5 (11)
Cases having hemorrhagic manifestations (%)	<50 (variable) (37)	34–83 (variable) (38)	40 (17, 29)	<1 (30)	3.6–6 (33)	<30 (variable) (34, 39)	N/A	12 (36)	0.5–5 (11)
Case fatality rate for cases with hemorrhagic/severe manifestations (%)	-	-	15–70 (17, 29)	<50 (30)	-	5–15 (7)	>50 (35)	30–60 (36) (40) (10)	Variable †
Outward signs of hemorrhagic disease	Maculopapular rash Petechiae Bleeding/bruising Bloody vomit/stool Bloody/red conjunctiva (3, 37)	See EBOV.	Bleeding from the mouth, nose, rectum, and vagina (17)	Macular rash Bruising Blood gums, stools, or vomit (21)	Petechiae Bruising Nosebleed Coughing up blood Bloody stools, vomit, or urine (33)	Petechiae Bruising Bloody conjunctiva Bloody stools, vomit, or urine (7, 20)	No outward signs of bleeding (7, 20)	Jaundice Liver and renal dysfunction Petechiae Bruising Bloody gums or vomit (9, 10)	Petechiae Bruising Bloody vomit/stool (16)
Approved vaccine available	Yes (41)	No (25)	No (42)	Only for livestock (14, 19)	No (6)	No (20)	No (20)	Yes (36) (9, 10)	No ‡

* The 2013–2016 outbreak was the largest on record (>28,000 cases).

** Many cases are asymptomatic, making true incidence difficult to estimate.

† Mortality for severe dengue can be up to 20 percent, but with proper supportive care, the mortality can be as low ~1 percent (11).

‡ There are 2 approved vaccines and 1 vaccine in phase 3 trials – however, the safety and efficacy of such vaccines is hotly debated (16, 43).

Abbreviations: EBOV – Ebola virus; MARV – Marburg virus; LASV – Lassa fever virus; RVFV – Rift Valley fever virus; CCHFV – Crimean Congo hemorrhagic fever virus; HTNV – Hantaan virus; SNV – sin nombre virus; YFV – yellow fever virus; DENV – dengue virus

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Table 3.

Hemostatic-related parameters and biomarkers in humans infected with different viruses causing viral hemorrhagic fever.

Parameter	EBOV	MARV	LASV	RVFV	CCHFV	HTNV	SNV	YFV	DENV
PT	Increased (56)	-	Increased *† (57–60)	Increased (61)	Increased * (62–66)	Normal ** (67, 68)	Increased (8, 15)	Increased (69, 70)	Normal (71–73)
aPTT	Increased (56)	Increased (74–76)	Increased † (60)	Increased (61)	Increased * (62–66, 77, 78)	Increased ** (67, 79)	Increased (8, 15)	Increased (70)	Increased (71–73)
TAT	-	-	Normal (80)	-	-	-	Increased (81) **	-	Increased (73, 82)
D-dimer/FDP	Increased (83–85) **	Increased (74)	Increased (86)	Increased (87, 88)	Increased * (62, 63, 66, 89, 90)	Increased ** (67, 79)	Increased (15, 81)	Increased (70, 91)	Increased (71, 73, 82)
Thrombocytopenia	Yes(56, 92)	Yes(74–76)	No * (57, 59, 60, 93)	Yes(61, 88, 94, 95)	Yes (62, 65, 66, 77, 78, 89, 90)	Yes ** (67, 68, 79)	Yes (15, 96) **	Yes (69, 70)	Yes (72, 73, 97)
TF	Increased (98) **	-	Increased (80)	-	-	-	Increased (81)	-	Increased (99–101)
tPA	Increased (84)	-	Increased (86)	Increased † (87)	-	-	Increased (102)	-	Increased (72, 73)
PAI-1	-	-	Increased (80, 86)	Normal † (87)	-	-	Increased (102, 103)	-	Increased (72, 73)
Elevated EC activation or EC glycocalyx disruption markers	s-TM s-ICAM-1 s-P-selectin (85)	-	s-TM s-ICAM-1 s-VCAM-1 s-vWF (80)	s-vWF s-VEGF (87)	s-ICAM-1 (104, 105) s-vWF (104)	-	s-VEGF (106)	Syndecan-1 (107)	Syndecan-1 (108)

* Minority of studies show normal values or opposite result

** Not all separated by viral strain/serotype

† Limited patient number

‡ Modest changes

Abbreviations: PT – prothrombin time; aPTT – activated partial thromboplastin time; TAT – thrombin anti-thrombin; FDP – fibrin degradation product; TF – tissue factor; tPA – tissue plasminogen activator; PAI-1 – plasminogen activator inhibitor-1; EC – endothelial cell; s – soluble; TM – thrombomodulin; vWF – von Willebrand factor; ICAM-1 – intercellular adhesion molecule-1; VEGF – vascular endothelial growth factor

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Table 4.

Hemostatic-related parameters and biomarkers in non-human primates infected with different viruses causing viral hemorrhagic fever.

Parameter	EBOV	MARV	LASV	RVFV	CCHFV	HTNV	SNV	YFV	DENV**
PT	Increased (109)	Increased (2)	Increased (110, 111)	Increased* (112–114)	-	-	Increased (115)	Increased (91, 116)	-
aPTT	Increased (109, 117)	Increased (2)	Increased (110, 111)	Increased (112–114)	Increased (118)	-	Increased (115)	Increased (116)	-
TAT	-	-	-	-	-	-	-	-	-
D-dimer/FDP	Increased (98, 117, 119, 120)	Increased (2, 121)	Increased (110, 111, 122)	Increased (112)	-	-	Normal (115)	Increased (91, 116)	-
Thrombocytopenia	Yes (98, 109, 117, 120, 123)	No* (2, 121, 124)	No* (110, 111, 122, 125, 126)	Yes (112, 114)	Yes (118, 127)	-	No (115)	Yes (116)	-
TF	Increased (98, 119, 128)	Increased (121)	-	-	-	-	-	-	-
tPA	Increased (98, 123)	-	-	-	-	-	-	-	-
PAI-1	Increased (123)	-	-	-	-	-	-	-	-

*Represents consensus of the majority of studies

**Does not cause overt clinical manifestations in non-human primates

Abbreviations: PT – prothrombin time; aPTT – activated partial thromboplastin time; TAT – thrombin anti-thrombin; FDP – fibrin degradation product; TF – tissue factor; tPA – tissue plasminogen activator; PAI-1 – plasminogen activator inhibitor-1;

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Table 5.

Disseminated intravascular coagulation scoring criteria according to the International Society on Thrombosis and Hemostasis.

Parameter	Values	Score
Platelet count ($\times 10^9$/L)	<50	2
	50–100	1
Prothrombin time (seconds)	Increased ≥6	2
	Increased 3–6	1
Fibrinogen (mg/dL)	<100	1
D-dimer	Strong increase	3
	Moderate increase	2